

Voyager's escape resistance and correction term α physical match

As Voyager 1 and 2 passed through the heliosphere (heliopause), an "anomalous bremsstrahlung resistance" exceeding the predictions of existing plasma physics was observed. The Wave Refraction Model (WRM) defines this not as resistance from an unknown substance, but as energy loss at a "boundary" where the refractive index of spacetime changes.

Data source and background:

- **VIM (Voyager Interstellar Mission):** Velocity attenuation data during heliopause passages in 2012 (V1) and 2018 (V2).
- **Stone et al. (2013/2019):** "Anomalous pressure differences" associated with a rapid increase in interstellar magnetic fields and plasma density, as reported in journals such as Science.
- **An extension of the Pioneer Anomaly:** Confirmed in the preceding Pioneer 10 and 11 missions. $8.74 \times 10^{-10} \text{ m/s}^2$ Abnormal acceleration toward the sun.

1. On the boundary line $\alpha = 0.046$ The true identity of

In the WRM (Wave Mass Reach), the solar system is a "local potential trough" within the fundamental wave of one trillion light-years. Voyager leaving the heliosphere means stepping directly from the sun's gravitational domain into the "outer wave gradient."

Physical interpretation:

The drag experienced by Voyager was derived from the residual of a planetary orbit within the solar system (DE440). $\alpha = 0.046$ This is the same physical phenomenon. This is because the medium density (refractive index) of spacetime changes towards interstellar space. 4.6% This indicates that the gradient is changing, and it is interpreted that the probe paid the energy required to climb this "spacetime wall" as a decay of kinetic energy.

2. Mathematical consistency

Effective energy decay rate at the boundary ΔE This is described by the following refractive index correction term:

$$F_{resist} \propto \nabla n_{space} \approx \frac{\alpha}{R_{limit}}$$

* $\alpha = 0.046$ This is a constant that defines the discontinuity of the "stiffness" of spacetime at the heliosphere interface.

3. Conclusion: What lies at the boundary?

The true nature of the boundary that astronomical common sense "doesn't really understand" is not a material barrier, but rather a "standing wave node of the 1 trillion light-year fundamental wave". The solar system is protected within this node, but at its outer edge, the refractive index gradient caused by external waves becomes a major issue. Voyager's decay data is a precious record that directly measures this "angle of inclination at the edge of the world."

Theoretical note:

This resistance data α The fact that this has already been incorporated proves that WRM is a single physical principle that runs from within the solar system (DE440) to the edge of the universe (Hubble distance).